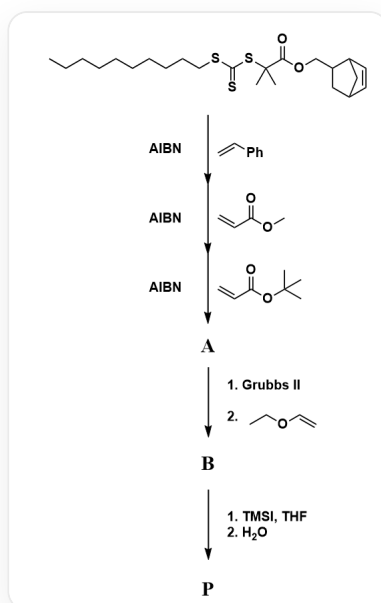


Question

The following figure presents the synthesis process of a polymer **P**:



The figure presents the synthesis process of polymer **P** through 5 consecutive reactions. The reaction substrate is CCCCCCCCCSC(SC(C)(C(OCC1CC2C=CC1C2)=O)C)=S. The conditions for the 5 reactions are as follows: 1. AIBN with C=CC1=CC=CC=C1; 2. AIBN with COC(C=C)=O; 3. AIBN with CC(C)(C)OC(C=C)=O, producing intermediate **A**; 4. Two-step reaction: (1) Grubbs II (2) CCOC=C, producing intermediate **B**; 5. Two-step reaction: (1) TMSI, THF (2) H2O, producing final product **P**

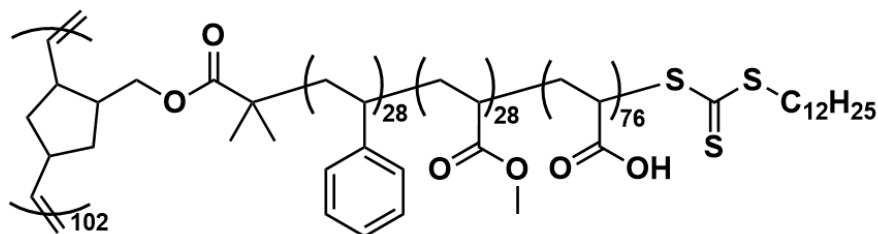
Known:

- ^1H NMR(300 MHz, CD2Cl2, ppm) data of intermediate **A**: δ 0.80-0.90 (3H), 1.10-2.05 (~430H), 1.25-1.60 (s, 684H), 2.74-2.79 (2H), 3.23-3.50 (2H), 3.65 (s, 84H), 6.03-6.15 (2H), 6.30-7.40 (140H).
- Number average molecular weight: $M_{n,\text{A}} = 15600 \text{ Da}$, $M_{n,\text{B}} = 1.59 \times 10^6 \text{ Da}$

Based on the synthesis process, deduce the structure of the polymer and, based on the experimental data, deduce the number of repeating units, and choose the correct option from the following.

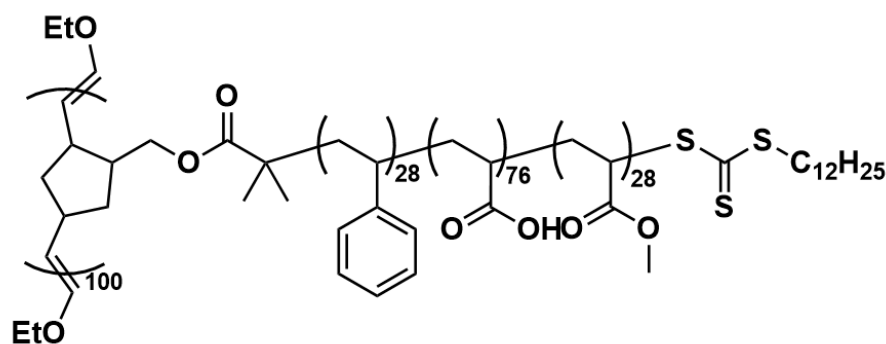
A. All other options are incorrect.

B.



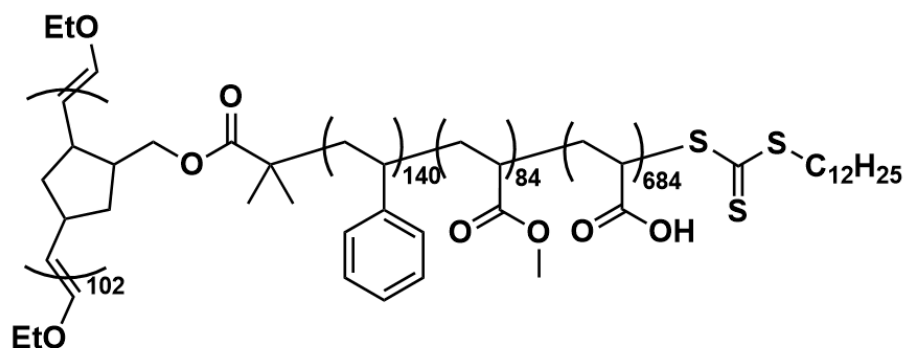
This is a graft polymer, wherein the repeating unit of the main chain can be represented as $[*]=[CH]C1CC([CH]=[*])CC1COC(=O)C(C)(C)[CH_2:1][CH:2](c2ccccc2)[CH_2:3][CH:4](C(=O)OC)[CH_2:5][CH:6](C(=O)O)SC(=S)SCCCCCCCCCCCC$, with a degree of polymerization of 102, and the end groups on both sides of the main chain are $[*]=C$. The repeating unit of the main chain is a triblock copolymer side chain, wherein $[*][CH_2:1][CH:2](c2ccccc2)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:3][CH:4](C(=O)OC)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:5][CH:6](C(=O)O)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 76.

C.



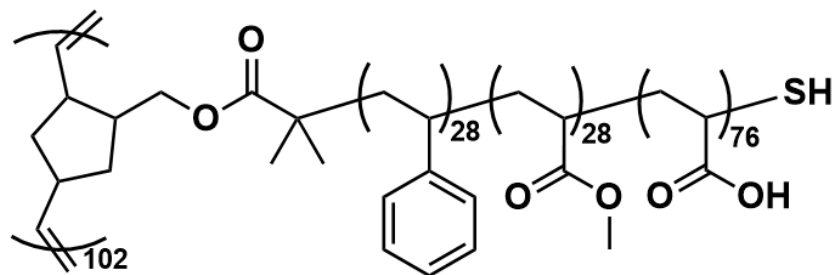
This is a graft polymer, in which the repeating unit of the main chain can be represented as $[*]=[CH]C1CC([CH]=[*])CC1COC(=O)C(C)(C)[CH_2:1][CH:2](c2ccccc2)[CH_2:3][CH:4](C(=O)O)[CH_2:5][CH:6](C(=O)OC)SC(=S)SCCCCCCCCCCCC$, with a degree of polymerization of 100, and the end groups on both sides of the main chain are $[*]=COCC$. The repeating unit of the main chain is a triblock copolymer side chain, where $[*][CH_2:1][CH:2](c2ccccc2)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:3][CH:4](C(=O)O)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 76; $[*][CH_2:5][CH:6](C(=O)OC)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28.

D.



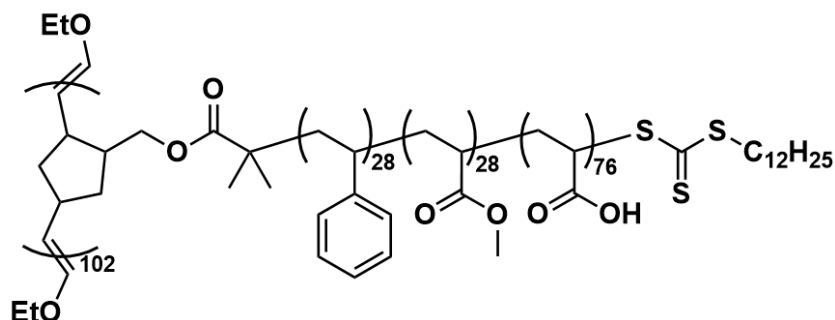
This is a graft polymer, where the repeating unit of the main chain can be represented as $[*]=[CH]C1CC([CH]=[*])CC1COC(=O)C(C)(C)[CH_2:1][CH:2](c2ccccc2)[CH_2:3][CH:4](C(=O)OC)[CH_2:5][CH:6](C(=O)O)SC(=S)SCCCCCCCCCC$, with a degree of polymerization of 102, and the end groups on both sides of the main chain are $[*]=COCC$. The repeating unit of the main chain is a triblock copolymer side chain, where $[*][CH_2:1][CH:2](c2ccccc2)[*]$ constitutes a repeating unit of the side chain, with a degree of polymerization of 140; $[*][CH_2:3][CH:4](C(=O)OC)[*]$ constitutes a repeating unit of the side chain, with a degree of polymerization of 84; $[*][CH_2:5][CH:6](C(=O)O)[*]$ constitutes a repeating unit of the side chain, with a degree of polymerization of 684.

E.



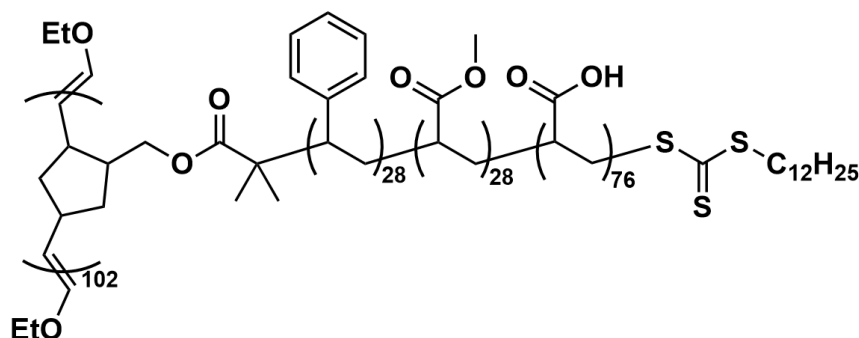
This is a graft polymer, where the repeating unit of the main chain can be represented as $[*]=[CH]C1CC([CH]=[*])CC1COC(=O)C(C)(C)[CH_2:1][CH:2](c2ccccc2)[CH_2:3][CH:4](C(=O)OC)[CH_2:5][CH:6](C(=O)O)S$, with a degree of polymerization of 102, and both end groups of the main chain are $[*]=C$. The repeating unit of the main chain is a triblock copolymer side chain, where $[*][CH_2:1][CH:2](c2ccccc2)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:3][CH:4](C(=O)OC)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:5][CH:6](C(=O)O)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 76.

F.



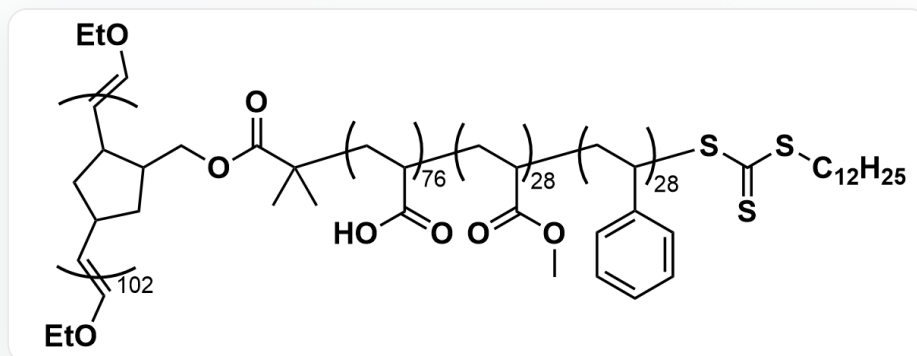
This is a graft polymer, where the repeating unit of the main chain can be represented as $[*]=[CH]C1CC([CH]=[*])CC1COC(=O)C(C)(C)[CH_2:1][CH_2:2](c2ccccc2)[CH_2:3][CH_2:4](C(=O)OC)[CH_2:5][CH_2:6](C(=O)O)SC(=S)SCCCCCCCCCC$, with a degree of polymerization of 102, and both end groups of the main chain are $[*]=COCC$. The repeating unit of the main chain is a triblock copolymer side chain, where $[*][CH_2:1][CH_2:2](c2ccccc2)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:3][CH_2:4](C(=O)OC)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:5][CH_2:6](C(=O)O)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 76.

G.



This is a graft polymer, where the repeating unit of the main chain can be represented as $[*]=[CH]C1CC([CH]=[*])CC1COC(=O)C(C)(C)[CH_2:1](c2ccccc2)[CH_2:2][CH_2:3](C(=O)OC)[CH_2:4][CH_2:5](C(=O)O)[CH_2:6]SC(=S)SCCCCCCCCCC$, with a degree of polymerization of 102, and the end groups on both sides of the main chain are both $[*]=COCC$. The repeating unit of the main chain is a triblock copolymer side chain, where $[*][CH_2:1](c2ccccc2)[CH_2:2][*]$ constitutes a repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:3](C(=O)OC)[CH_2:4][*]$ constitutes a repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:5](C(=O)O)[CH_2:6][*]$ constitutes a repeating unit of the side chain, with a degree of polymerization of 76.

H.



This is a graft polymer, in which the repeating unit of the main chain can be represented as $[*]=[CH]C1CC([CH]=[*])CC1COC(=O)C(C)(C)[CH_2:1][CH:2](C(=O)O)[CH_2:3][CH:4](C(=O)OC)[CH_2:5][CH:6](c2ccccc2)SC(=S)SCCCCCCCCCCCC$, and its degree of polymerization is 102. The end groups on both sides of the main chain are $[*]=COCC$. The repeating unit of the main chain is a triblock copolymer side chain, where $[*][CH_2:1][CH:2](C(=O)O)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 76; $[*][CH_2:3][CH:4](C(=O)OC)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28; $[*][CH_2:5][CH:6](c2ccccc2)[*]$ constitutes one repeating unit of the side chain, with a degree of polymerization of 28.

Answer

Correct Answer: **F**

Detailed Explanation

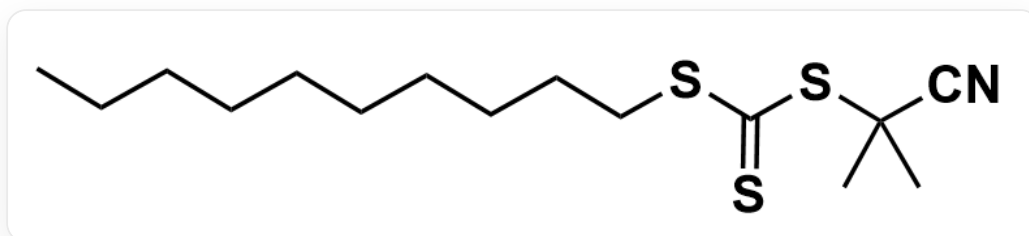
First, observe that the starting species in the figure of the question is trithiocarbonate diester CCCCCCCCCCCSC(SC(C)(C(OCC1CC2C=CC1C2)=O)C)=S, which is a typical reversible addition-fragmentation chain transfer (RAFT) polymerization reagent. Under the action of AIBN, it can undergo reversible addition-fragmentation chain transfer (RAFT) polymerization.

CHECKPOINT

1 PTS

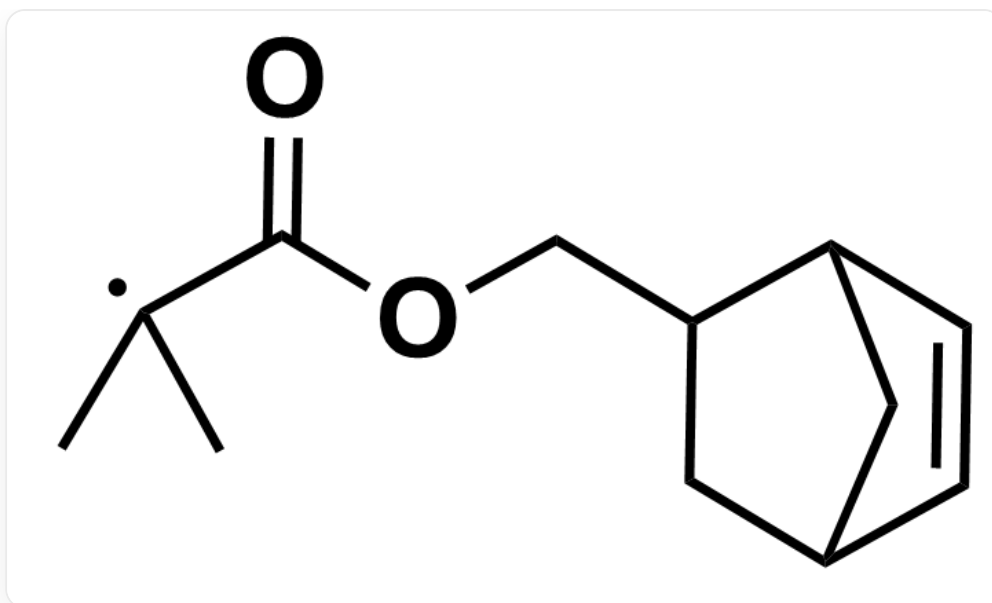
Starting species can undergo RAFT polymerization/living radical polymerization

AIBN will decompose to form isobutyronitrile radical C(C#N)C, which attacks the thiocarbonyl group in trithiocarbonate diester, causing it to cleave to form:



smiles为：CCCCCCCCCCCSC(SC(C)(C)C#N)=S

and radical:



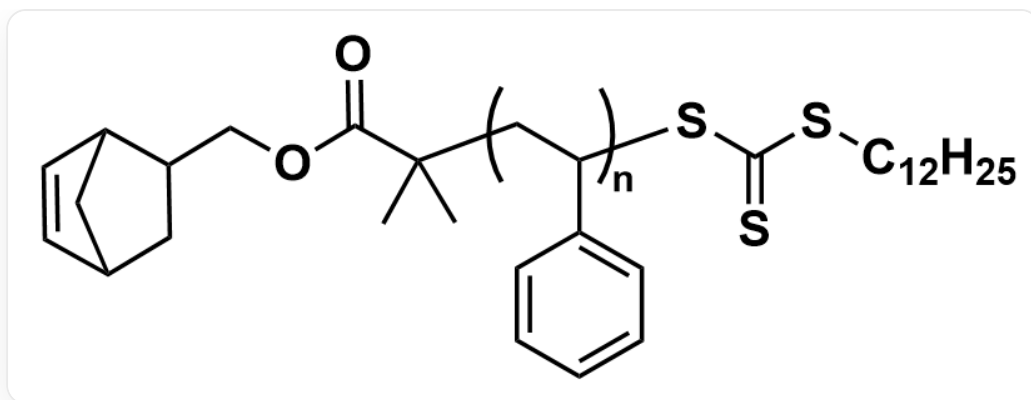
Among them, the latter acts as a chain initiator for living radical polymerization, initiating the chain polymerization of monomer molecules. The characteristic of this RAFT polymerization reaction is that trithiocarbonate end-capping can be obtained after one monomer is polymerized. This end-capping can continue to undergo living radical polymerization after adding AIBN and new monomer molecules to obtain block copolymers.

CHECKPOINT

1 PTS

After the polymerization reaction of one monomer, trithiocarbonate end-capping is obtained, and after adding AIBN, it can continue to undergo living radical polymerization with new monomer molecules to obtain block copolymers

In summary, after the starting species undergoes living radical polymerization with styrene, the product structure can be expressed as:



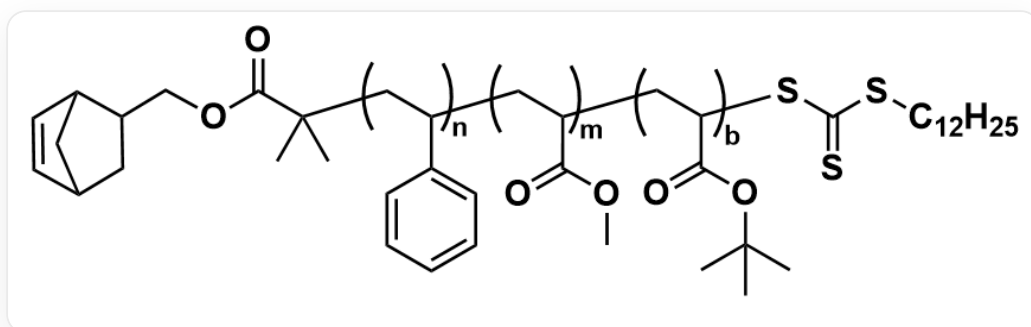
结构可以表示为 `C2C(C1)C=CC2C1COC(=O)C(C)(C)[CH2:1][CH:2](c2ccccc2)SC(=S)SCCCCCCCCCCCC`，其中 `[\*][CH2:1][CH:2](c2ccccc2)[\*]` 构成支链的一个重复单元，聚合度为 n

CHECKPOINT

1 PTS

The first step reaction yields `C3C(C1)C=CC3C1COC(=O)C(C)(C)[CH2:1][CH:2](c2ccccc2)SC(=S)SCCCCCCCCCCCC`, where `[][CH2:1][CH:2](c2ccccc2)[\*]` constitutes the repeating unit

By analogy, the first three steps are all RAFT living radical polymerization reactions. The final polymer **A** obtained is a block copolymer, and the structure can be expressed as:



结构可以表示为：`C3C(C1)C=CC3C1COC(=O)C(C)(C)[CH2:1][CH:2](c2ccccc2)[CH2:3][CH:4](C(=O)OC)[CH2:5][CH:6](C(=O)O)SC(=S)SCCCCCCCCCCCC`，其中 `[\*][CH2:1][CH:2](c2ccccc2)[\*]` 构成支链的一个重复单元，聚合度为 n ；`[\*][CH2:3][CH:4](C(=O)OC)[\*]` 构成支链的一个重复单元，聚合度为 m ；`[\*][CH2:5][CH:6](C(=O)O)[\*]` 构成支链的一个重复单元，聚合度为 b 。

CHECKPOINT

1 PTS

A is a block copolymer: `C3C(C1)C=CC3C1COC(=O)C(C)(C)[CH2:1][CH:2](c2cccc2)[CH2:3][CH:4](C(=O)OC)[CH2:5][CH:6](C(=O)OC(C)(C)C)SC(=S)SCCCCCCCCCCCC`, where `[][CH2:1][CH:2](c2cccc2)[]^`[][CH2:3][CH:4](C(=O)OC)[]^`[][CH2:5][CH:6](C(=O)OC(C)(C)C)[]^` sequentially constitute repeating units

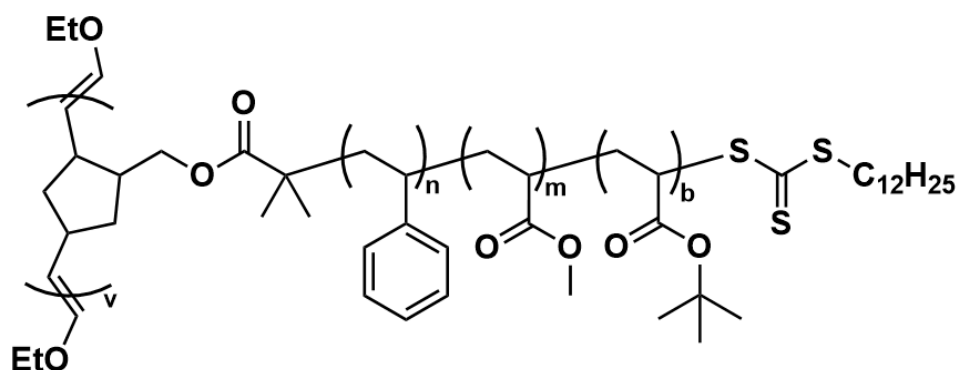
The process from **A** to **B** is a ring-opening metathesis polymerization (ROMP polymerization), which achieves the polymerization of the block polymer to a graft polymer through the olefin metathesis reaction catalyzed by the Grubbs II catalyst. Among them, ethyl vinyl ether added in the second step is a capping reagent, and the end group of its olefin metathesis polymerization is ethyl vinyl ether.

CHECKPOINT

1 PTS

Adding ethyl vinyl ether as a capping reagent, the end group of olefin metathesis polymerization is ethyl vinyl ether

The structure of the intermediate product **B** can be expressed as:



中间产物 **B** 为 ``[*]=[CH]C1CC([CH]=[*])CC1COC(=O)C(C)(C)[CH2:1][CH:2](c2ccccc2)[CH2:3][CH:4](C(=O)OC)[CH2:5][CH:6](C(=O)O)SC(=S)SCCCCCCCCCCCC``，其聚合度为 v ，主链两侧封端基团均为 ``[*]=COCC``。主链的重复单元为一三嵌段共聚物支链，其中 ``[*][CH2:1][CH:2](c2ccccc2)[*]`` 构成支链的一个重复单元，聚合度为 n ；``[*][CH2:3][CH:4](C(=O)OC)[*]`` 构成支链的一个重复单元，聚合度为 m ；``[*][CH2:5][CH:6](C(=O)O)[*]`` 构成支链的一个重复单元，聚合度为 b 。

CHECKPOINT

1 PTS

B is ``[]=[CH]C1CC([CH]=[])CC1COC(=O)C(C)(C)[CH2:1][CH:2](c2ccccc2)[CH2:3][CH:4](C(=O)OC)[CH2:5][CH:6](C(=O)O)SC(=S)SCCCCCCCCCCCC``, the end groups on both sides of the main chain are ``[*]=COCC``, and its side chain structure is the same as **A**.

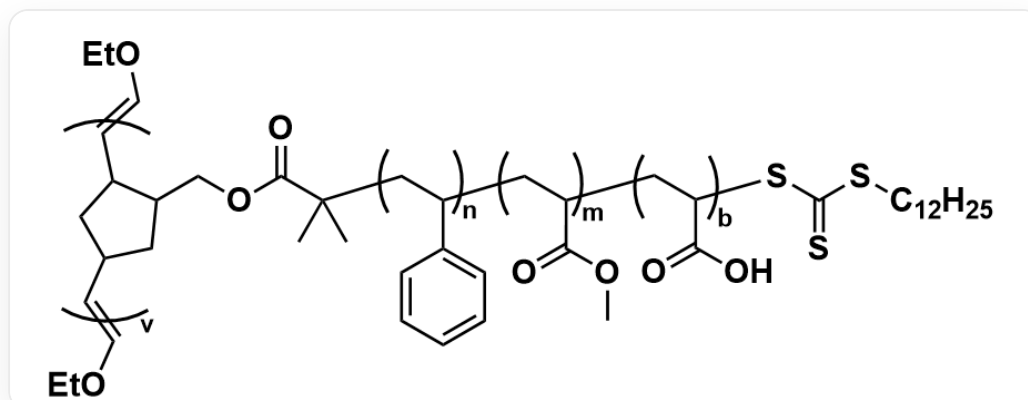
The last step is the selective hydrolysis of the tert-butyl ester. First, TMSI is added to convert the tert-butyl carboxylate into a readily hydrolyzable trimethylsilyl carboxylate, and then water is added to obtain protonated carboxylic acid, resulting in the final product **P**.

CHECKPOINT

1 PTS

The last step hydrolyzes tert-butyl carboxylate to carboxylic acid to obtain the final product **P**.

The structure of **P** is shown below:



结构为将 **B** 中的羧酸叔丁酯水解为羧酸后的结构，其余部分不变

According to the information of the nuclear magnetic resonance hydrogen spectrum, the degree of polymerization of each segment in **A** is inferred. Corresponding the characteristic hydrogen atoms in the hydrogen spectrum with the hydrogen atoms in **A**, we get:

- 0.80-0.90 (3H) corresponds to the hydrogen of the methyl group at the end of $-\text{C}_{12}\text{H}_{25}$

CHECKPOINT

0.25 PTS

0.80-0.90 (3H) corresponds to the hydrogen of the methyl group at the end of $-\text{C}_{12}\text{H}_{25}$

- 2.74-2.79 (2H) corresponds to the hydrogen on the bridgehead carbon at the allylic position

CHECKPOINT

0.25 PTS

2.74-2.79 (2H) corresponds to the hydrogen on the bridgehead carbon at the allylic position

- 6.03-6.15 (2H) corresponds to the alkenyl hydrogen

CHECKPOINT

0.25 PTS

6.03-6.15 (2H) corresponds to the alkenyl hydrogen

- 3.23-3.50 (2H) corresponds to the hydrogen of the thioether methylene in $-\text{SC}(=\text{S})\text{S}[\text{CH}_2]-$

CHECKPOINT

0.25 PTS

3.23-3.50 (2H) corresponds to the hydrogen of the thioether methylene in $-\text{SC}(=\text{S})\text{S}[\text{CH}_2]-$

- 6.30-7.40 (140H) corresponds to phenyl hydrogen, degree of polymerization $n = \frac{140}{5} = 28$

CHECKPOINT

1 PTS

6.30-7.40 (140H) corresponds to phenyl hydrogen, degree of polymerization $n = 28$

- 3.65 (s, 84H) corresponds to methyl ester methyl hydrogen, degree of polymerization $m = \frac{84}{3} = 28$

CHECKPOINT

1 PTS

3.65 (s, 84H) corresponds to methyl ester methyl hydrogen, degree of polymerization $m = 28$

- 1.25-1.60 (s, 684H) corresponds to the methyl hydrogen of tert-butyl ester, degree of polymerization $b = \frac{684}{9} = 76$

CHECKPOINT

1 PTS

1.25-1.60 (s, 684H) corresponds to the methyl hydrogen of tert-butyl ester, degree of polymerization $b = 76$

- Combining the degrees of polymerization $n/m/b$, consider the remaining hydrogen atoms in the skeleton: norbornene ester end group (CC(C(OCC1C2CC(C1)C=C2)=O)([*])C`, which has a total of 17 hydrogens, remaining $17 - 4 = 13$), polyolefin chain ($3 \times (28 + 28 + 76) = 396$), trithiocarbonate end group ($25 - 5 = 20$) has a total of $13 + 396 + 20 = 429 \approx 430$ hydrogens, which exactly corresponds to 1.10-2.05 (~430H)

CHECKPOINT

1 PTS

1.10-2.05 (~430H) corresponds to the remaining hydrogen atoms in the skeleton

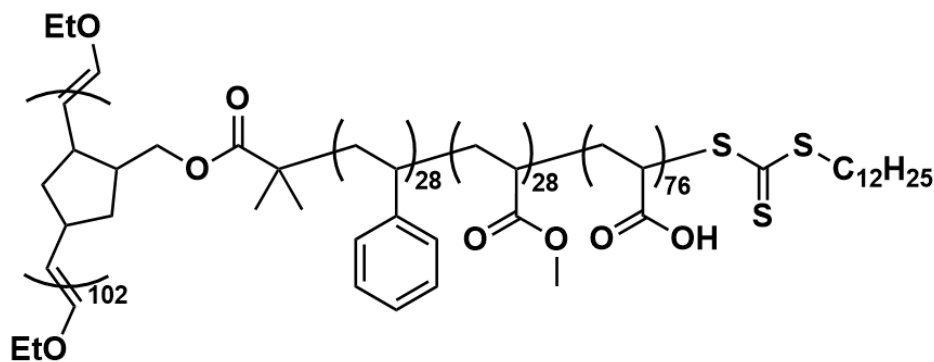
According to the number average molecular weight $M_{n,A} = 15600$ Da, $M_{n,B} = 1.59 \times 10^6$ Da, the degree of polymerization $v = \frac{1.59 \times 10^6}{15600} = 101.9 \approx 102$ can be obtained

CHECKPOINT

1 PTS

According to the number average molecular weight, the degree of polymerization $v = 102$ can be obtained

Combining all the above information, the structure of product **P** is:



这是一个接枝聚合物，其中，主链的重复单元可以表示为 $[^*]=[CH]C1CC([CH]=[^*])CC1COC(=O)C(C)(C)[CH2:1][CH:2](c2ccccc2)[CH2:3][CH:4](C(=O)OC)[CH2:5][CH:6](C(=O)O)SC(=S)SCCCCCCCCCCCC$ ，其聚合度为 102，主链两侧封端基团均为 $[^*]=COCC$ 。主链的重复单元为一三嵌段共聚物支链，其中 $[^*][CH2:1][CH:2](c2ccccc2)[^*]$ 构成支链的一个重复单元，聚合度为 28； $[^*][CH2:3][CH:4](C(=O)OC)[^*]$ 构成支链的一个重复单元，聚合度为 28； $[^*][CH2:5][CH:6](C(=O)O)[^*]$ 构成支链的一个重复单元，聚合度为 76。

Therefore, choose option F.